

Baker's yeast β -glucan supplementation increases monocytes and cytokines post-exercise: implications for infection risk?

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Strenuous aerobic exercise is known to weaken the immune system, and while many nutritional supplements have been proposed to boost post-exercise immunity, few are known to be effective. The purpose of the present study was to evaluate whether 10 d of supplementation with a defined source of baker's yeast β -glucan (BG, Wellmune WGP®) could minimise post-exercise immunosuppression. Recreationally active men and women (n 60) completed two 10 d trial conditions using a cross-over design with a 7 d washout period: placebo (rice flour) and baker's yeast BG (250 mg/d of β -1,3/1,6-glucans derived from *Saccharomyces cerevisiae*) before a bout of cycling (49 ± 6 min) in a hot ($38 \pm 2^\circ\text{C}$), humid (45 ± 2 % relative humidity) environment. Blood was collected at baseline (before supplement), pre- (PRE), post- (POST) and 2 h (2H) post-exercise. Total and subset monocyte concentration was measured by four-colour flow cytometry. Plasma cytokine levels and lipopolysaccharide (LPS)-stimulated cytokine production were measured using separate multiplex assays. Total (CD14⁺) and pro-inflammatory monocyte concentrations (CD14⁺/CD16⁺) were significantly greater at POST and 2H ($P < 0.05$) with BG supplementation. BG supplementation boosted LPS-stimulated production of IL-2, IL-4, IL-5 and interferon- γ (IFN- γ) at PRE and POST ($P < 0.05$). Plasma IL-4, IL-5 and IFN- γ concentrations were greater at 2H following BG supplementation. It appears that 10 d of supplementation with BG increased the potential of blood leucocytes for the production of IL-2, IL-4, IL-5 and IFN- γ . The key findings of the present study demonstrate that BG may have potential to alter immunity following a strenuous exercise session.